SSS Criterion of similarity

If three sides of a triangle are proportional to the three corresponding sides of another triangle, then the two triangles are similar.

So if, $\frac{AB}{PQ} = \frac{AC}{PR} = \frac{BC}{QR}$ then $\Delta ABC \sim \Delta PQR$

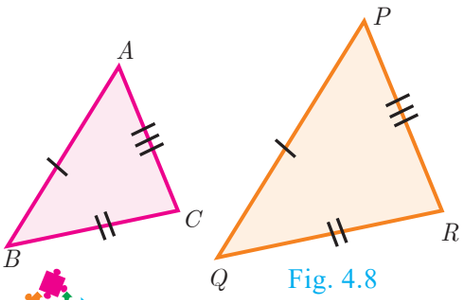


Fig. 4.8

Thinking Corner

Are any two right angled triangles similar? If so why?

Some useful results on similar triangles

1. A perpendicular line drawn from the vertex of a right angled triangle divides the triangle into two triangles similar to each other and also to original triangle.

   $\Delta ADB \sim \Delta BDC$, $\Delta ABC \sim \Delta ADB$, $\Delta ABC \sim \Delta BDC$

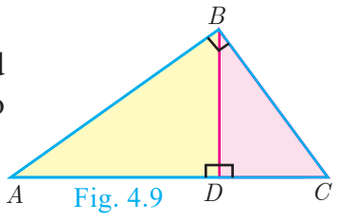


Fig. 4.9

2. If two triangles are similar, then the ratio of the corresponding sides are equal to the ratio of their corresponding altitudes.

   i.e. if $\Delta ABC \sim \Delta PQR$ then

   $\frac{AB}{PQ} = \frac{BC}{QR} = \frac{CA}{RP} = \frac{AD}{PS} = \frac{BE}{QT} = \frac{CF}{RU}$

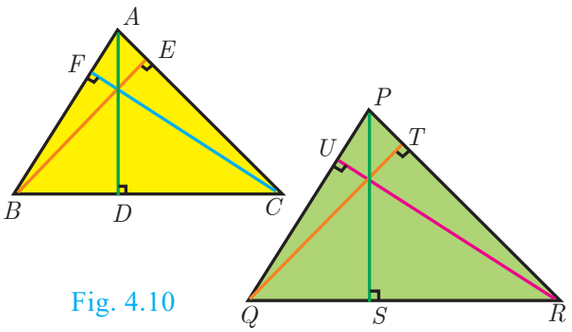


Fig. 4.10

3. If two triangles are similar, then the ratio of the corresponding sides are equal to the ratio of the corresponding perimeters.

   $\Delta ABC \sim \Delta DEF$ then

   $\frac{AB}{DE} = \frac{BC}{EF} = \frac{CA}{FD} = \frac{AB + BC + CA}{DE + EF + FD}$

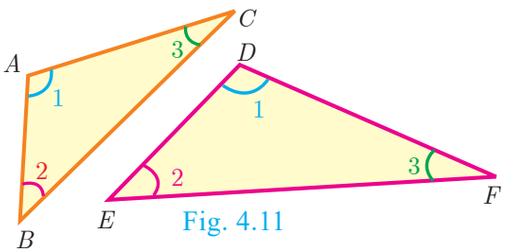


Fig. 4.11

4. The ratio of the area of two similar triangles are equal to the ratio of the squares of their corresponding sides.

   $\frac{\text{area}(\Delta ABC)}{\text{area}(\Delta PQR)} = \frac{AB^2}{PQ^2} = \frac{BC^2}{QR^2} = \frac{AC^2}{PR^2}$

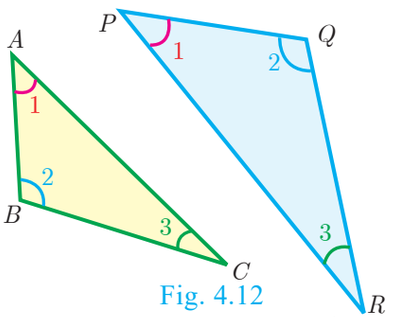


Fig. 4.12

5. If two triangles have common vertex and their bases are on the same straight line, the ratio between their areas is equal to the ratio between the length of their bases.

   Here, $\frac{\text{area}(\Delta ABD)}{\text{area}(\Delta BDC)} = \frac{AD}{DC}$.

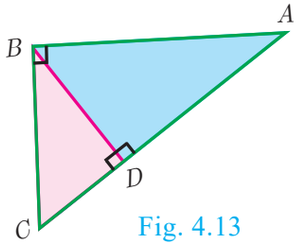


Fig. 4.13

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